ENGR 111
Introduction to Engineering
Class #9

Unit conversions and dimensional analysis

The number is only half the answer

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

Where we are today

The design process has eight steps. Two of them are where unit errors get in.

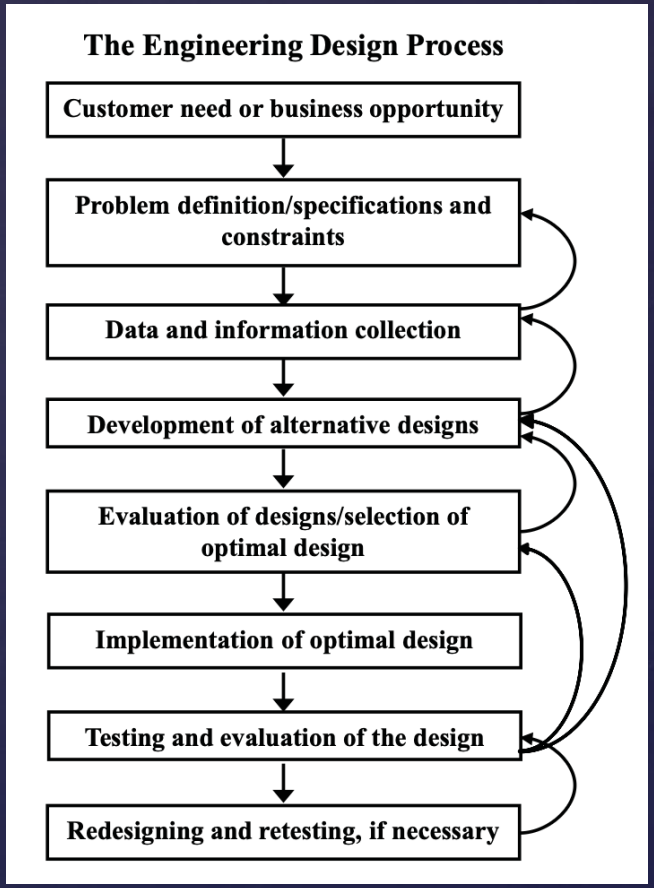
Step 2 writes numbers down. A spec is a number and a unit.

Step 3 involves collecting data. You will be collecting component specs, performance data, measurements, etc. (all are numbers with units) from datasheets, standards, measurements, and other teams.

Watch the units. The rest of the world finds "feet" to be rather funny.

Step 2 ➡

Step 3 ➡



When a number arrives from a supplier or another team, check the unit is the one you assumed.

A $125 million unit conversion mistake

September 23, 1999. The Mars Climate Orbiter was lost



Spacecraft (orbiter) build by Lockheed Martin

Sends telemetry data back to earth

→

Lockheed Martin ground software

Calculates thruster impulse
Outputs **lbf·s**

→

JPL Navigation Software

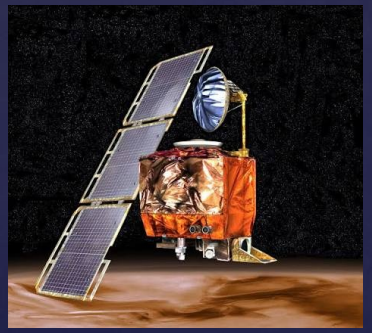
Expects **N·s**

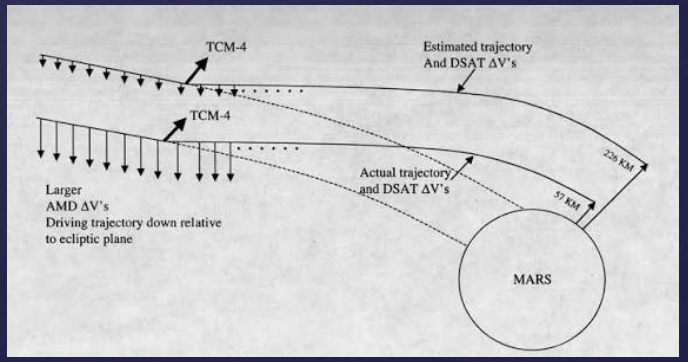
→

Underestimates every thruster correction by 4.45×

1 lbf·s ≈ 4.45 N·s (A factor 4.45 mismatch)

The interface document specified newton seconds. The check that would have caught it was not run.





Seven base units, everything else built

Quantity	Base unit	Quantity	Base unit
length	meter, m	temperature	kelvin, K
mass	kilogram, kg	amount of substance	mole, mol
time	second, s	luminous intensity	candela, cd
electric current	ampere, A		

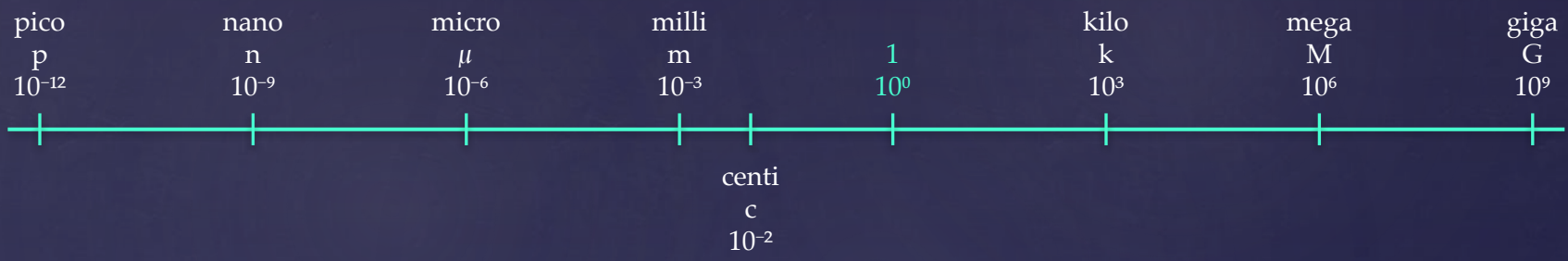
A measurement is a comparison against an agreed standard. Say “three” and the question is three of what.

Since 2019 all seven are defined by constants of nature, not by objects. The kilogram used to be a metal cylinder in a vault outside Paris. It is now fixed by the Planck constant.

So a lab in Oregon and a lab in Tokyo can each build their own kilogram from the same constant, instead of borrowing the cylinder.

The prefixes

The nine on your prefix handout. These are the ones worth memorizing.



Case matters. M is mega, m is milli. A factor of a billion between them.

Every one of them is a power of ten, so a prefix conversion is only ever moving a decimal point.

Centi is the odd one. The rest step by a thousand each time. Centi steps by a hundred, which is why it sits so close to milli.

Worked. 5 μg of carbon nanotubes is 5×10^{-6} g, which is 0.005×10^{-3} mg, which is 0.005 mg.

Watch μ. On a keyboard it usually gets typed as "u", so "um" in a datasheet means micrometers.

Derived units

Named units that are combinations of the seven base units.

Quantity	Unit	In base units
force	newton, N	$kg \cdot m / s^2$
energy	joule, J	$N \cdot m = kg \cdot m^2 / s^2$
power	watt, W	J / s
pressure	pascal, Pa	N / m^2
charge	coulomb, C	$A \cdot s$
potential difference	volt, V	W / A
resistance	ohm, Ω	V / A

Each named unit is a shorthand. A volt is a watt per amp, which is a joule per coulomb. The name saves you writing $kg \cdot m^2 / (s^3 \cdot A)$ every time.

Watts and watt-hours measure different things. One is a rate, the other an amount. A 7 kW charger running two hours delivers 14 kWh.

When you meet an unfamiliar unit, write it out in base units.

The pound problem

In US customary units, "pound" means two different quantities.

	POUND-FORCE · lbf	**POUND-MASS · lbm**
What it measures	A force.	A mass.
In SI	1 lbf = 4.448 N	1 lbm = 0.4536 kg
Where you meet it	What a scale pushes back with, what a bolt is torqued to, what a thruster produces.	What is actually inside the bag of flour.
Its counterpart	The matching mass unit is the slug: 1 slug = 32.174 lbm.	The matching force unit is the pound-force.

1. At standard Earth gravity, 1 lbm weighs 1 lbf. That numerical coincidence is part of what makes "pounds" confusing.

2. Write lbf or lbm rather than just "lb" when somebody else has to read it.

Converting Units Step by Step

A conversion factor tells you how one unit relates to another.

Example: Convert gallons to liters using quarts as an intermediate unit..

$$18\ \text{gal} \times \frac{4\ \text{qt}}{1\ \text{gal}} \times \frac{1\ \text{L}}{1.05\ \text{qt}} = 68.6\ \text{L}$$

Write the units into the fraction every time, not just the numbers.

Cancel on paper. If the units do not cancel down to the one you want, the arithmetic does not matter.

Set the whole chain up before touching a calculator. One expression, one evaluation.

Units that cancel to the one you wanted are decent evidence the setup is right. But obviously they will not catch a wrong conversion factor.

Squared and cubed units

The conversion factor takes the same exponent as the unit.

150 ft² to yd²

$$150\ \text{ft}^2 \times \left(\frac{1\ \text{yd}}{3\ \text{ft}}\right)^2 = \frac{150\ \text{yd}^2}{9} = 16.7\ \text{yd}^2$$

Wrong: divide by 3.

Dimensional analysis

- In engineering you will be writing down equations. Dimensional analysis is a powerful and simple method to check if you are on the right track.
- It is simply checking that the dimensions (units) are the same on the left-hand and right-hand side of an equation.
- We usually write the units with backets [ ]
- Example 1:

  Suppose you derive this equation (x = position, a = acceleration, t = time)

  $x = (1/2)at^2$

  $[\mathrm{m}] = [\mathrm{m/s^2}]\,[\mathrm{s^2}] = [\mathrm{m}]$ ✓

- Example 2

  Suppose you derive this equation (x = position, a = acceleration, t = time)

  $x = (1/2)at$

  $[\mathrm{m}] = [\mathrm{m/s^2}]\,[\mathrm{s}] = [\mathrm{m/s}]$ ×

Units that work do not prove the equation is true, but units that fail prove it is wrong.

Checking Schrodinger equation via dimensional analysis

$$-(\hbar^2 / 2m)\cdot d^2\psi/dx^2 + V(x)\,\psi = E$$

$$\Rightarrow \quad \frac{[\mathrm{J\cdot s}][\mathrm{J\cdot s}]}{[\mathrm{kg}]}\cdot\frac{1}{[\mathrm{m^2}]}\,\psi + [\mathrm{J}] = [\mathrm{J}]$$

$$\Rightarrow \quad [\mathrm{J^2}]\cdot\underbrace{\frac{[\mathrm{s^2}]}{[\mathrm{kg\cdot m^2}]}}_{=\,1/[\mathrm{J}]} + [\mathrm{J}] = [\mathrm{J}] \quad \checkmark$$

Can you show this?
Hint Use this equation:
Work = Force × distance

What you need to know. $\hbar$ is Planck's constant divided by 2π: $\hbar = 1.055 \times 10^{-34}$ joule-seconds. m is a mass in kilograms, x a position in meters, V and E are energies. ψ is the wavefunction.

You do not need the unit of ψ. It appears once in every term, so whatever its unit is, the same factor is on both sides.

The derivative just divides. d^2/dx^2 means "per meter, twice", so that term carries ψ divided by meters squared.

The other two terms are an energy times ψ. All three come out as joules times ψ. It passes.

Worth trying on any formula somebody hands you that you do not recognize.